

Daniel Karnofel

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github.com/danielkarnofel

Education

California State University, Monterey Bay <i>B.S. in Computer Science (In Progress)</i>	09.2024 - 08.2026 GPA: 4.00
City College of San Francisco <i>Coursework in Computer Science</i>	01.2023 - 06.2024 GPA: 4.00
University of California, Santa Barbara <i>B.A. in Philosophy, Minor in Art and Technology</i>	09.2017 - 06.2021 GPA: 3.19

Experience

California State University, Monterey Bay <i>Teaching Assistant - Internet Programming</i> Graded assignments, provided code reviews, and assisted students with full-stack web application development using JavaScript , HTML , CSS , Node , Express , and MySQL .	10.2025 - Present
The Document Foundation <i>Open Source Contributor - Libre Office</i> Implemented C++ codebase improvements and created documentation for the LibreOffice open source software suite.	10.2025 - Present
UCSF Medical Center <i>Practice Coordinator</i> Created and maintained 100+ pages of reference and training material, led process improvement efforts by tracking, categorizing, and coordinating workflow changes, and served as primary consultant for data analysis and visualization projects. Completely eliminated the occurrence of non-trivial policy errors for the entire team within 6 months.	04.2023 - Present

Projects

2D Rendering Engine

A full-featured, general purpose 2D rendering engine built from scratch with **JavaScript** and **WebGL**. Includes a full UI with customizable canvas attributes, 3 primitive object types with dynamic transformation, scale, and rotation, as well as 3 pre-built test scenes.

Solar System Simulation

A 3D solar system simulation built from scratch with **JavaScript** and **WebGL**. Includes a full 3D scene-graph system, custom shaders for Phong/Gouraud/flat shading, and support for skyboxes, flow mapping, and more.

Image Processing Library

A software based image processing library built with **C++**. Includes a wide variety of basic per-pixel color and tone operations, geometric operations like cropping and resizing, and image composition operations like blending, multiplying, alpha-masking, and more. Also supports application of predefined or custom convolution kernels.

File Compression Tool

A file compression command line tool built with **C++** that uses LZ77 frequency analysis and Huffman coding to provide lossless data compression. Includes convenient output statistics and a custom binary compression format.

Technical Skills

C, C++, Java, Android Studio, HTML, CSS, JavaScript, Node, Express, SQL, WebGL, OpenGL